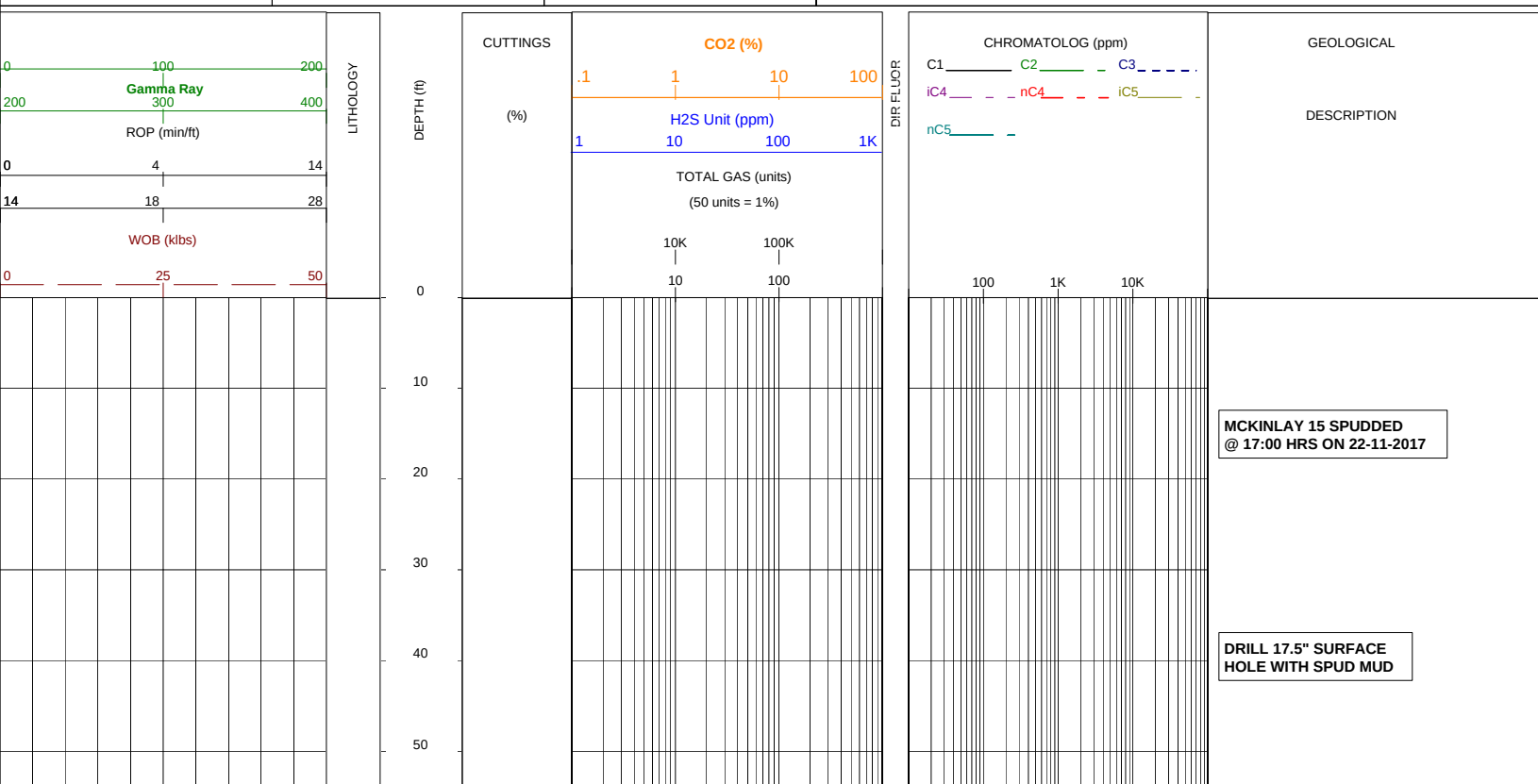
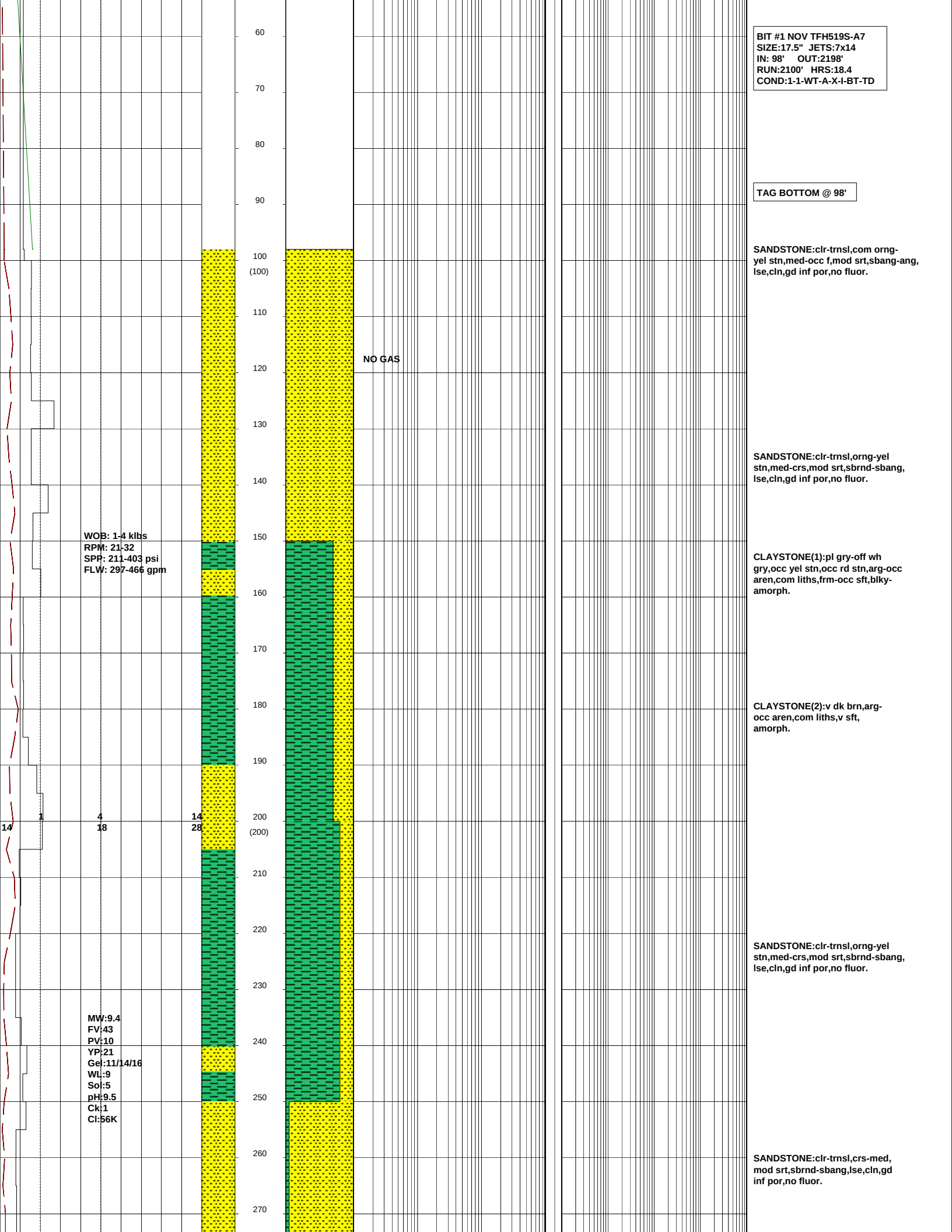
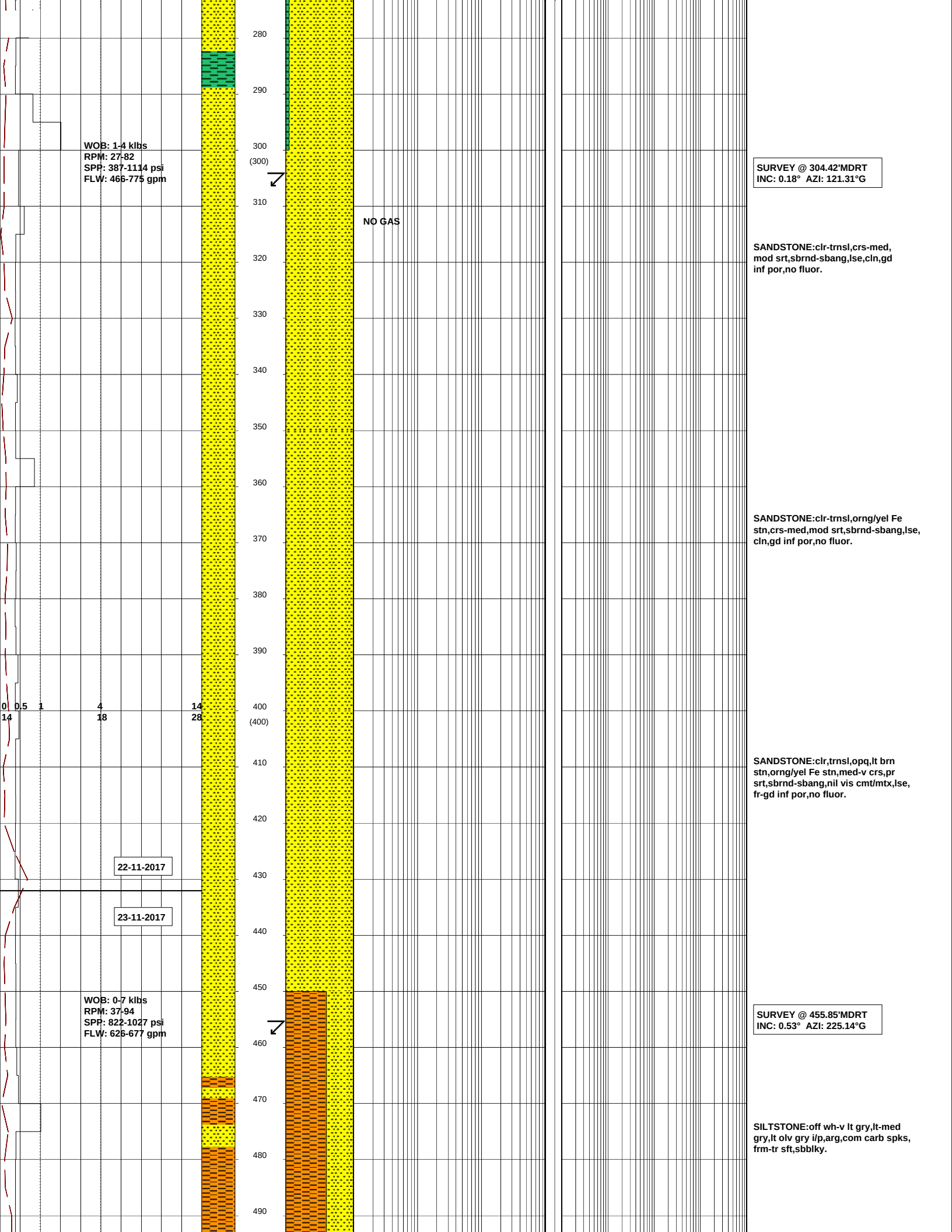


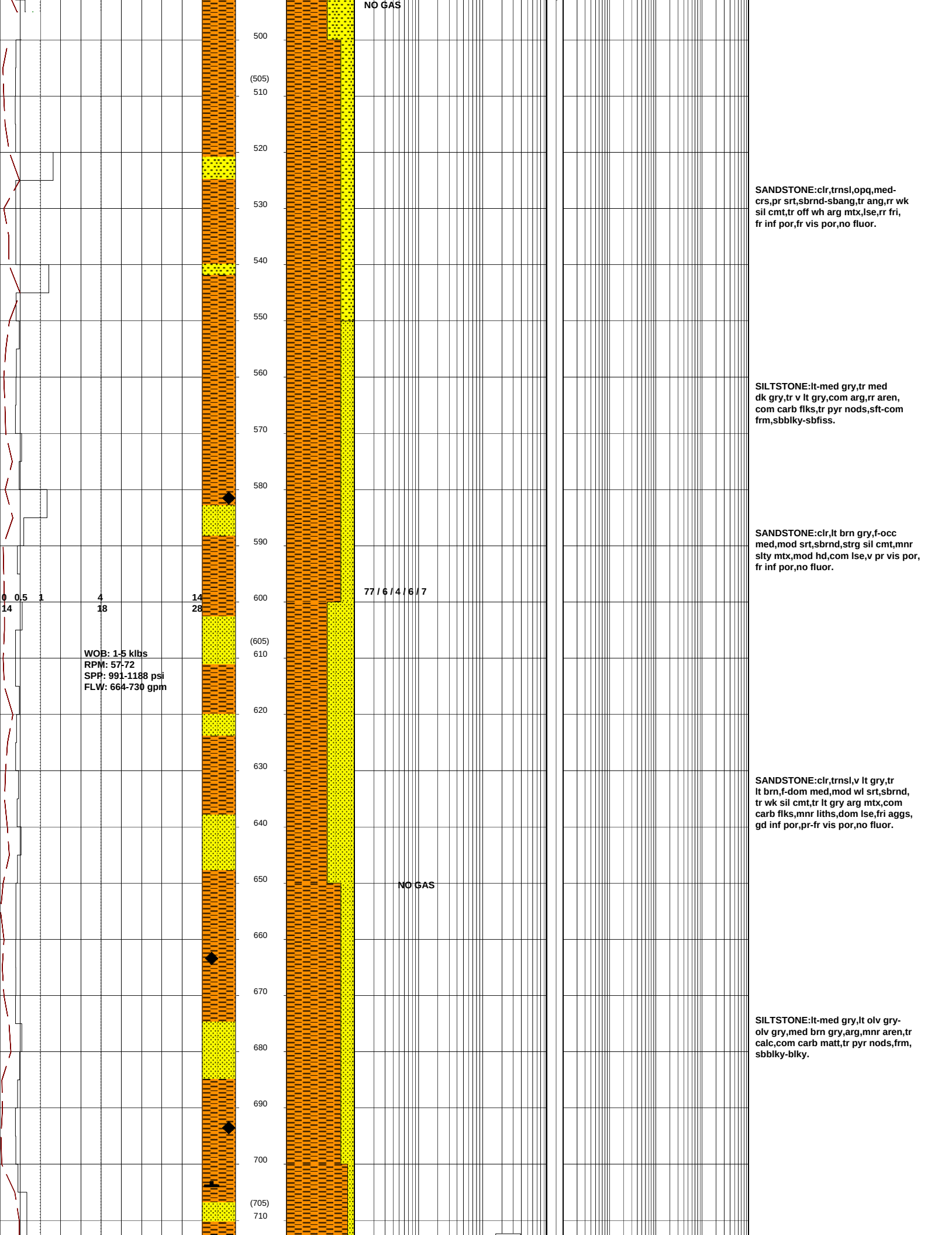
Field : MCKINLAY	Rotary Table : 150.0'	Rig : ENSIGN 971	Open Hole:	Cased Hole:	Loggers :	PHILIP LUU
Permit : PPL234	Ground Level : 126.5'	Spud Date : 22-11-2017	17.50" 2198'	13.375" 2188'		SEAN KETTLE
State : SOUTH AUSTRALIA	GDA 94 Co-ordinates :	TD Date : 02-12-2017	12.25" 5045'	9.625" 5036'		VAHID JALEH
Country : AUSTRALIA	Lat. : 28°28'19.10"S	Total Depth : 8495'MD 4092.1'TVD	8.5" 8495'			
Scale : 1/240 5"=100'	Long. : 140°29'28.99"E	Final Status : BAREFOOT COMPLETION				

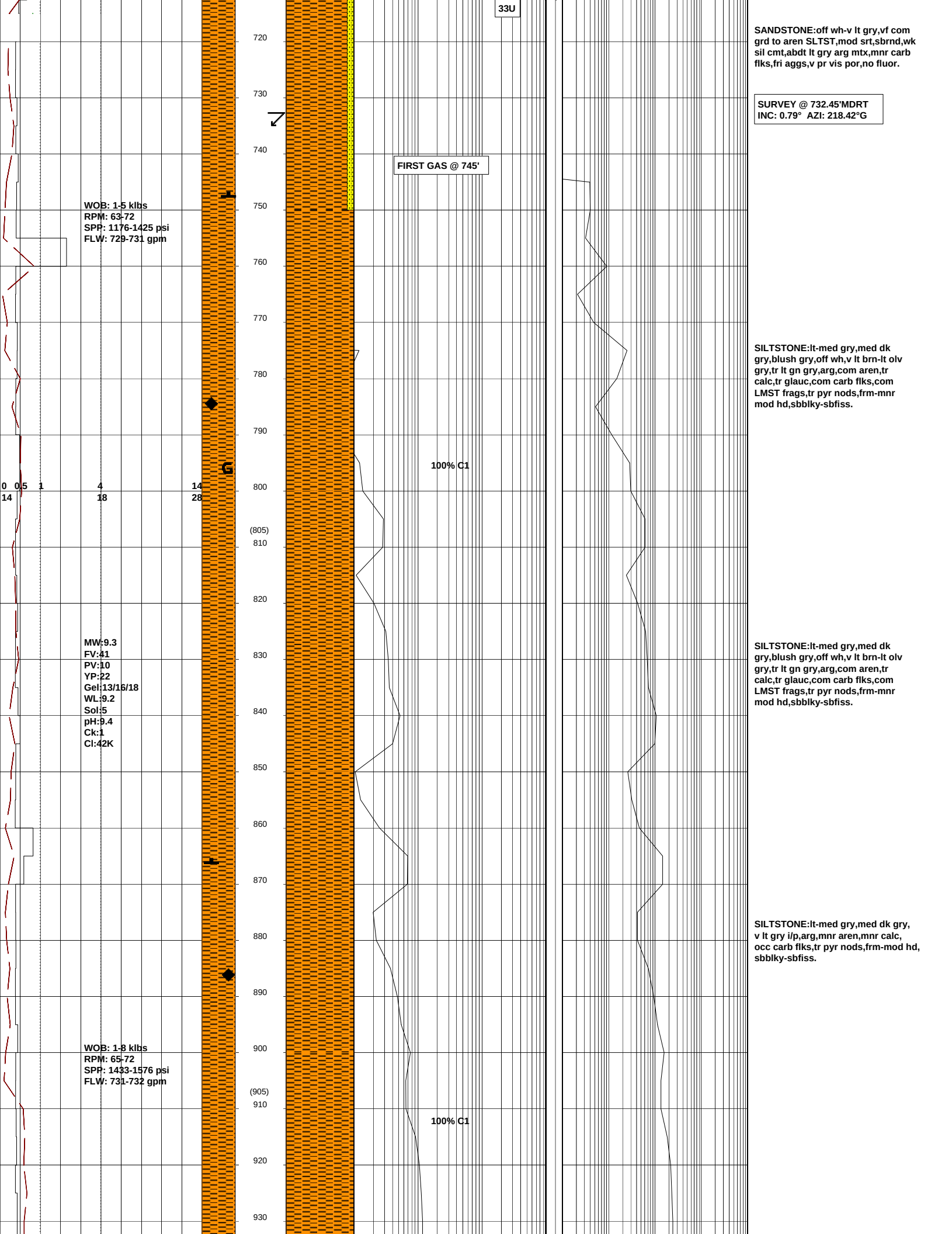
LITHOLOGY	ACCESSORIES	DRILLING DATA	ABBREVIATIONS
 Coarse sandstone	 Pyrite	 Casing shoe	BOPD - Barrels of Oil Per Day
 Medium sandstone	 Siderite	 Bit Trip	OG - Over Gauge
 Fine sandstone	 Glauconite	 Wiper Trip	BWPD - Barrels of Water Per Day
 Cement	 Iron	 Core	CG - Connection Gas
 Siltstone	 Mica	 Dst	CO - Circulate Out
 Carb Siltstone	 Chert	 Survey	COND - Condensate
 Metamorphic rock	 Calcareous		c/c - Crush Cut
 Claystone	 Dolomitic		DST - Drill Stem Test
 Limestone	 Carbonaceous		FLOW - Flow Rate (gal/min)
 Coal-lignite	 Fossils		GCM - Gas Cut Mud
			GCW - Gas Cut Water
			GTS - Gas To Surface
			INJ - Injection of Mist (bbls/hr)
			LCM - Lost Circulation Material
			MMCFD - Million Cubic Feet / Day
			NGTS - No Gas To Surface
			NOTS - No Oil To Surface
			NR - No Returns
			OCM - Oil Cut Mud
			OG - Over Gauge
			OH - Open Hole
			OTS - Oil To Surface
			Q - Flow Rate
			REC - Recovery
			Rmf - Resistivity mud filtrate
			ROP - Rate Of Penetration
			RPM - Revolutions Per Minute
			RTSTM - Rate Too Small To Measure
			Rw - Resistivity water
			r/r - Ring Residue
			SCFM - Standard Cubic Ft/Min (air)
			SGCM - Slightly Gas Cut Mud
			SPM - Strokes Per Minute
			SPP - Stand Pipe Pressure
			SWC - Side-Wall Core
			TG - Trip Gas
			WOB - Weight On Bit

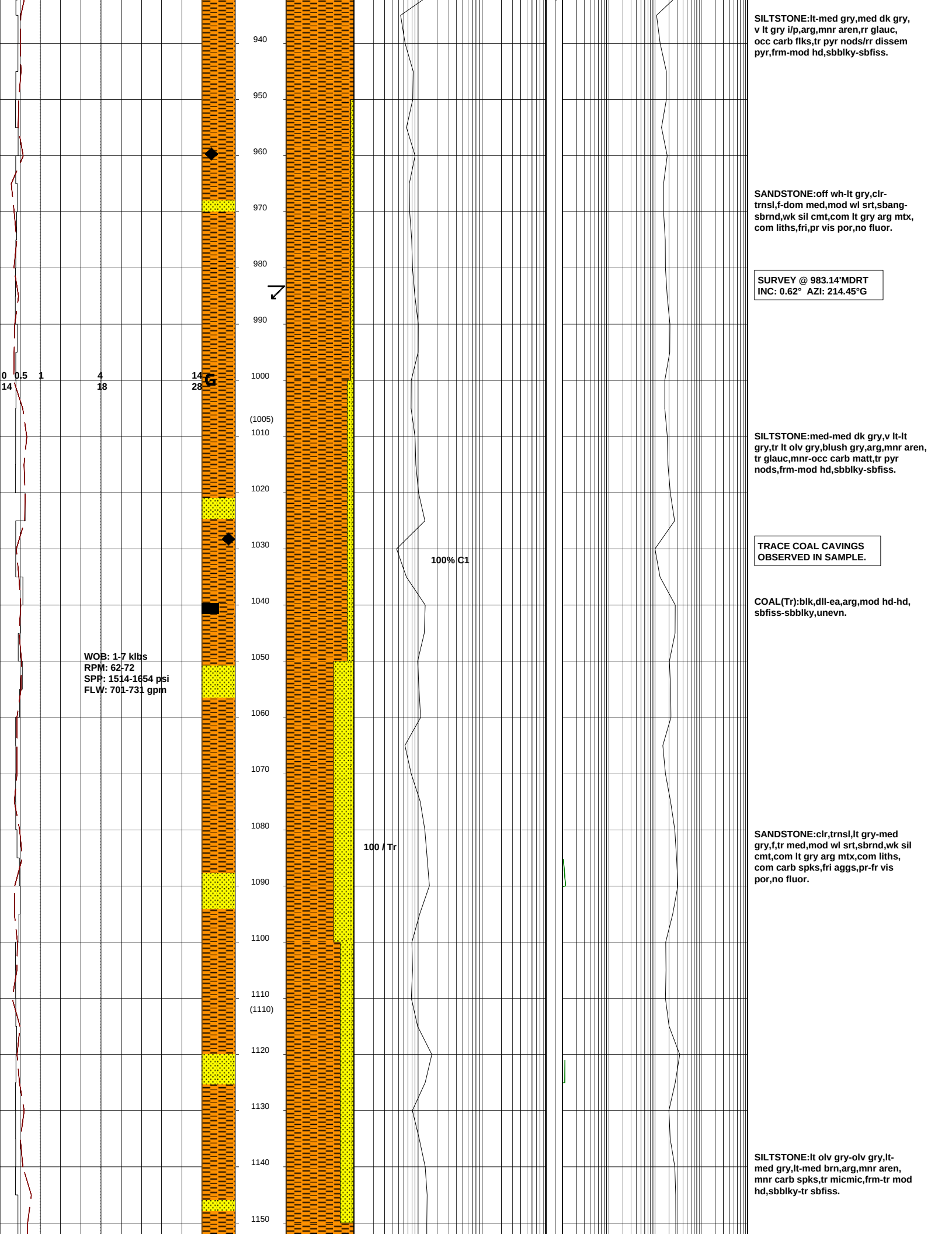


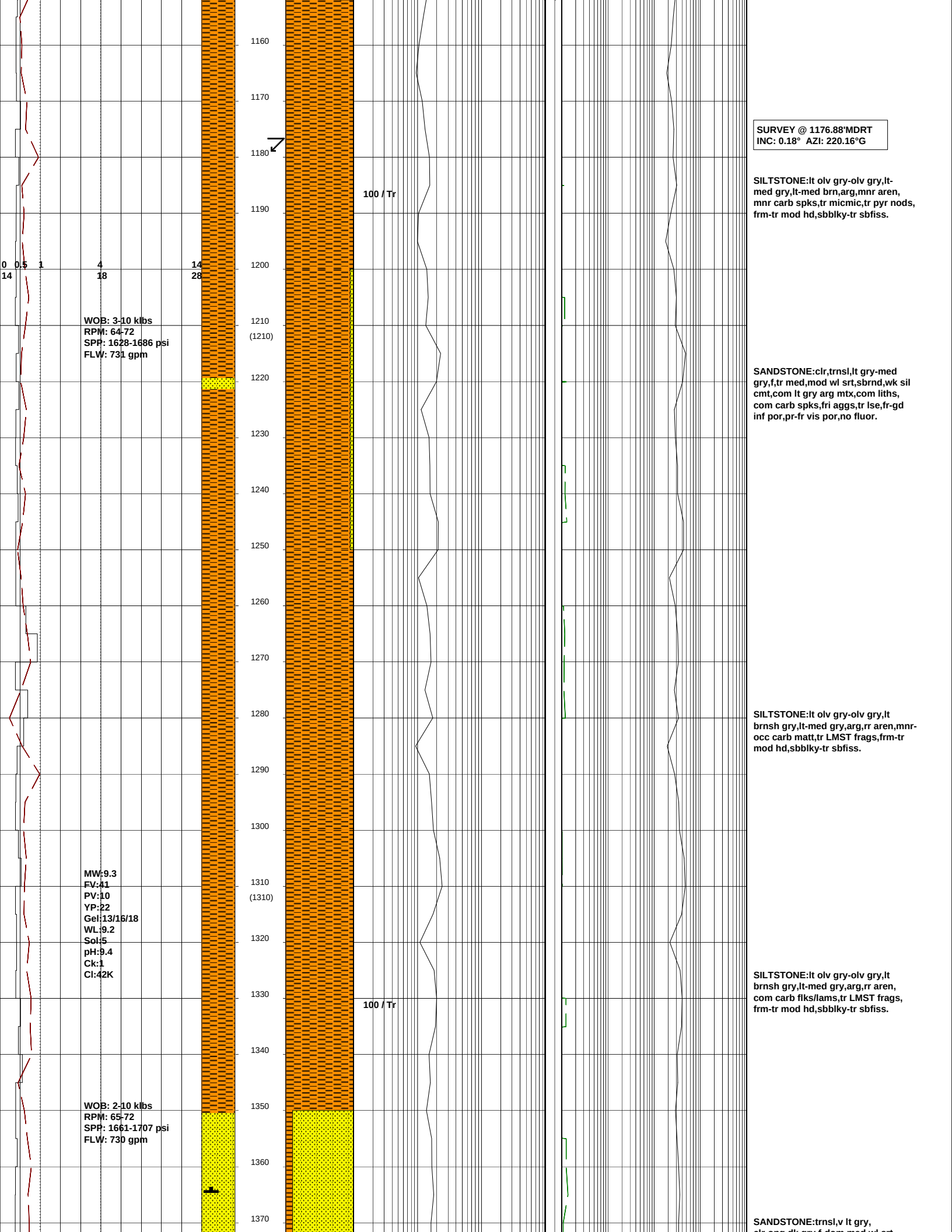












SURVEY @ 1176.88'MDRT
INC: 0.18° AZI: 220.16°G

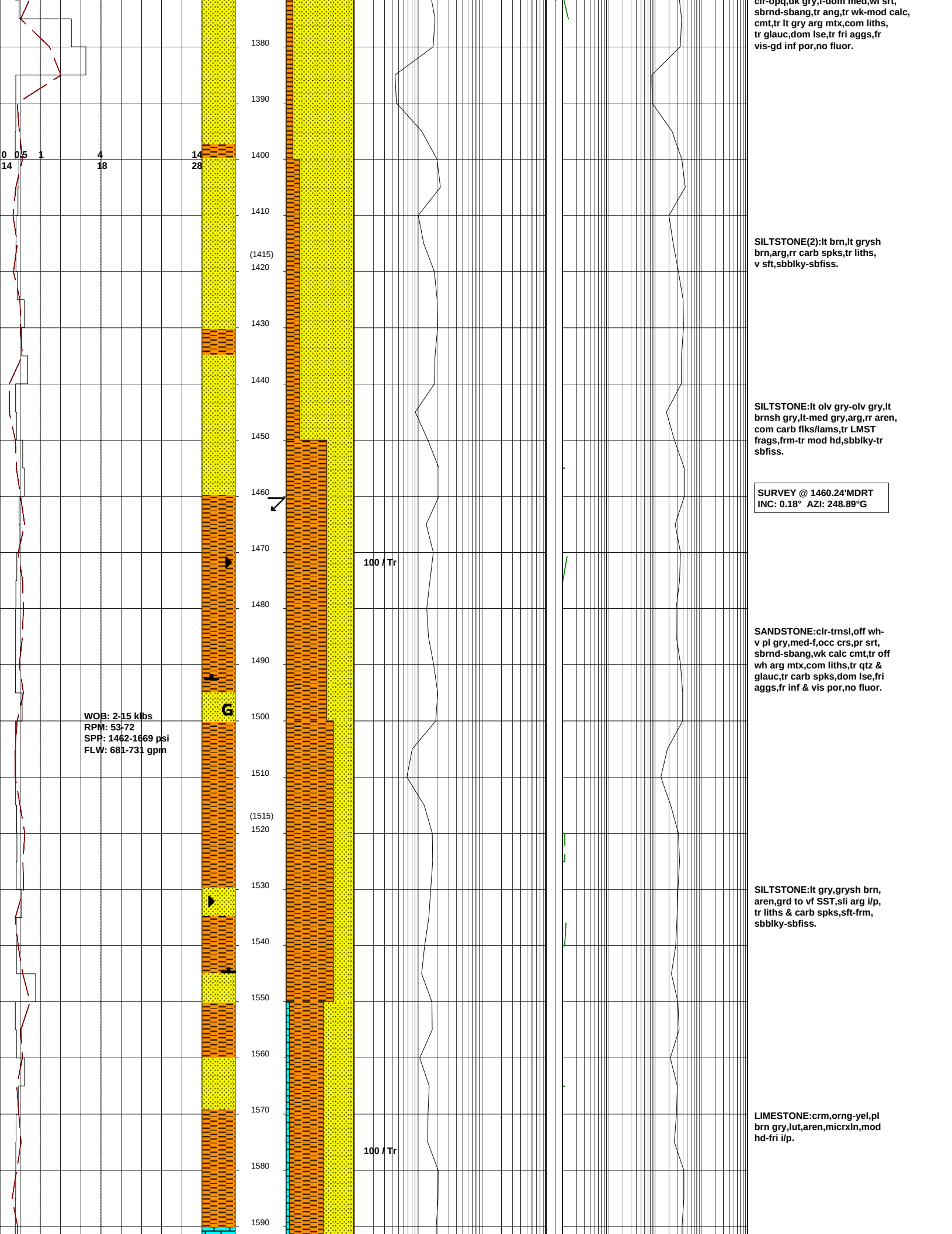
SILTSTONE:lt olv gry-olv gry,lt-med gry,lt-med brn,arg,mnr aren,mnr carb spks,tr micmic,tr pyr nods,frm-tr mod hd,sbbiky-tr sbfiss.

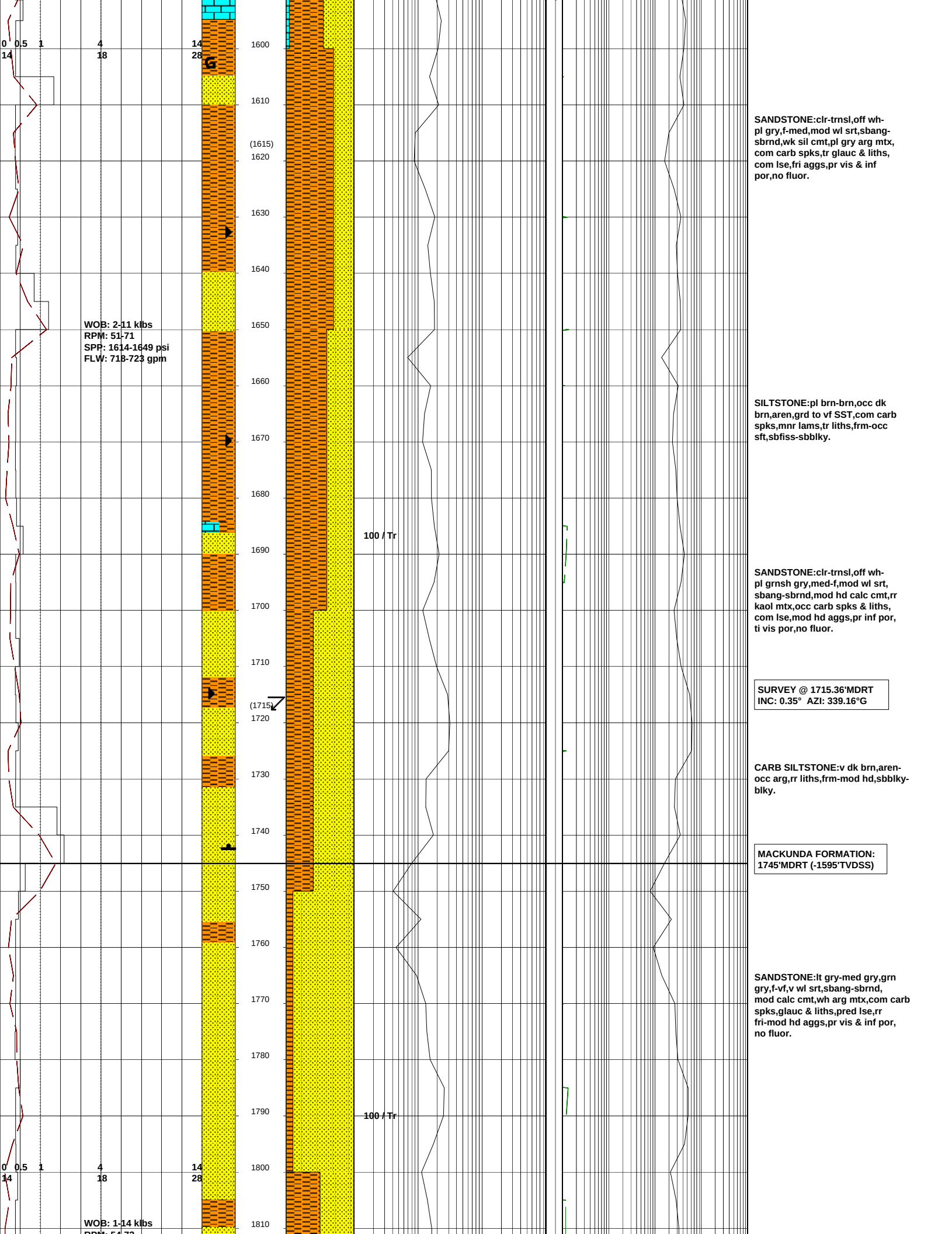
SANDSTONE:clr,trnsl,lt gry-med gry,f,tr med,mod wl srt,sbrnd,wk sil cmt,com lt gry arg mtx,com liths,com carb spks,fri aggs,tr lse,fr-gd inf por,pr-fr vis por,no fluor.

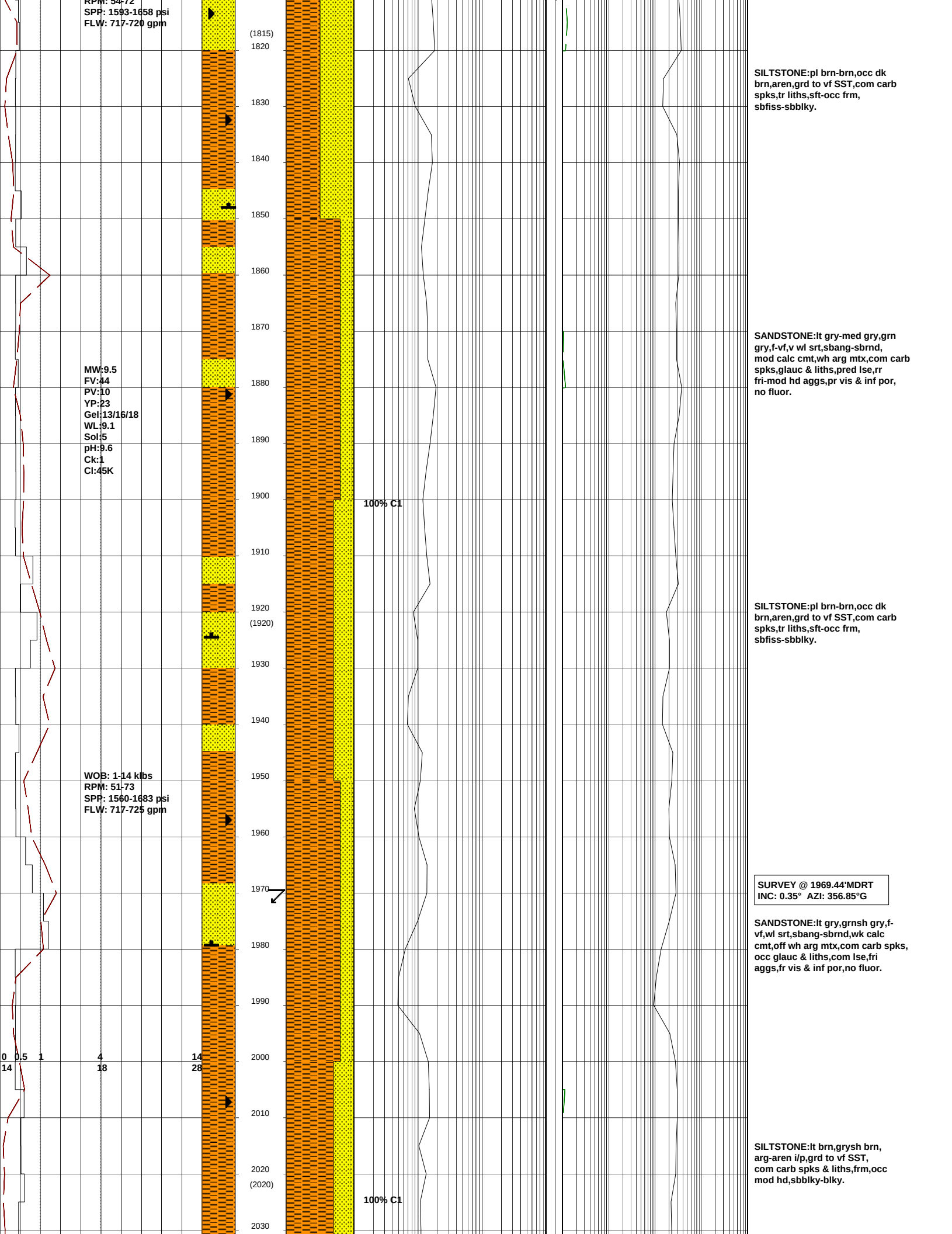
SILTSTONE:lt olv gry-olv gry,lt brnsh gry,lt-med gry,arg,rr aren,mnr-occ carb matt,tr LMST frags,frm-tr mod hd,sbbiky-tr sbfiss.

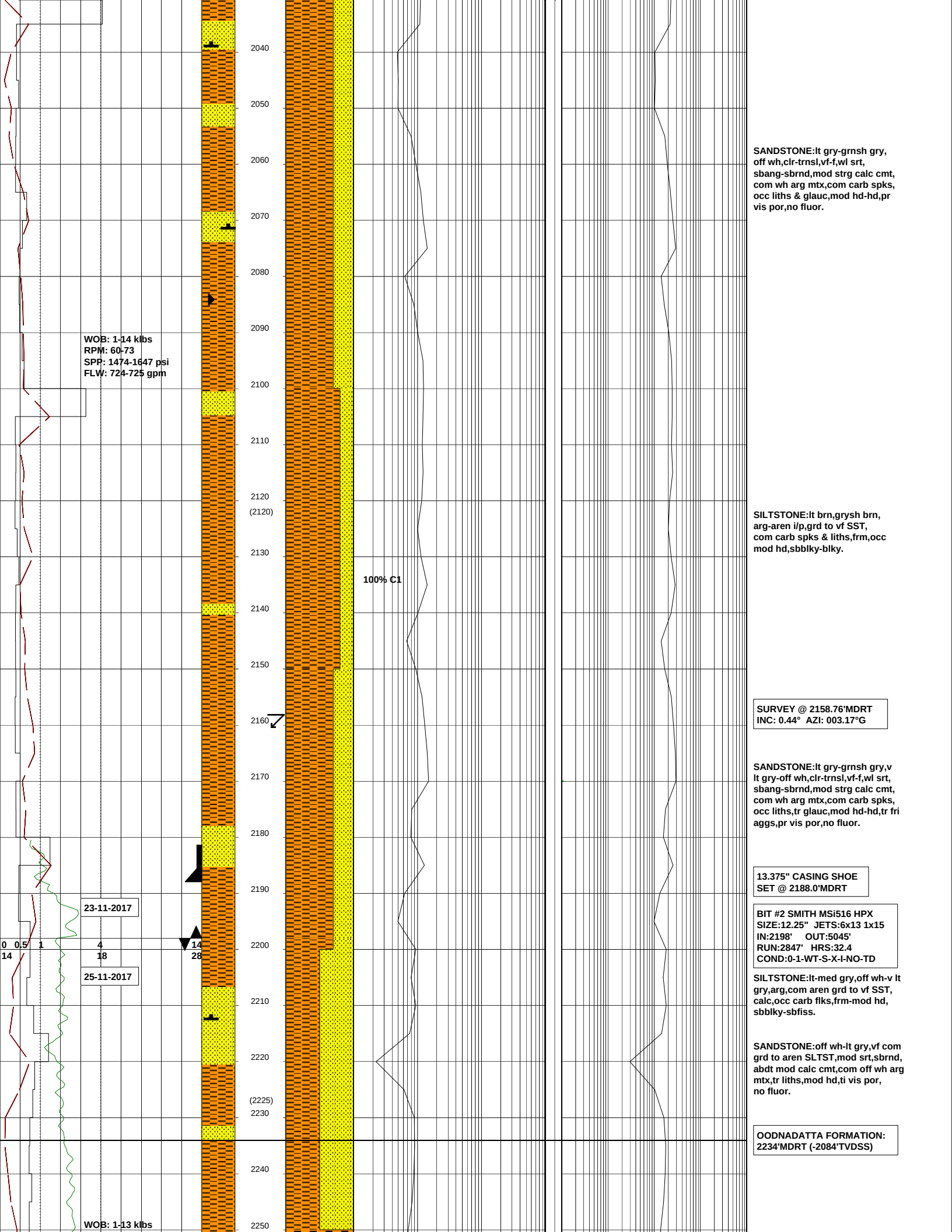
SILTSTONE:lt olv gry-olv gry,lt brnsh gry,lt-med gry,arg,rr aren,com carb flks/lams,tr LMST frags,frm-tr mod hd,sbbiky-tr sbfiss.

SANDSTONE:trnsl,v lt gry,clr,med,mod wl srt,sbrnd,wk sil cmt,com lt gry arg mtx,com liths,com carb spks,fri aggs,tr lse,fr-gd inf por,pr-fr vis por,no fluor.









WOB: 1-14 kips
RPM: 60-73
SPP: 1474-1647 psi
FLW: 724-725 gpm

SANDSTONE:lt gry-grnsh gry,
off wh,clr-trnsl,vf-f,wl srt,
sbang-sbrnd,mod strg calc cmt,
com wh arg mtx,com carb spks,
occ liths & glauc,mod hd-hd,pr
vis por,no fluor.

SILTSTONE:lt brn,grysh brn,
arg-aren ip,grd to vf SST,
com carb spks & liths,frm,occ
mod hd,sbblky-blky.

SURVEY @ 2158.76'MDRT
INC: 0.44° AZI: 003.17°G

SANDSTONE:lt gry-grnsh gry,v
lt gry-off wh,clr-trnsl,vf-f,wl srt,
sbang-sbrnd,mod strg calc cmt,
com wh arg mtx,com carb spks,
occ liths,tr glauc,mod hd-hd,tr fri
aggs,pr vis por,no fluor.

13.375" CASING SHOE
SET @ 2188.0'MDRT

BIT #2 SMITH MSi516 HPX
SIZE:12.25" JETS:6x13 1x15
IN:2198' OUT:5045'
RUN:2847' HRS:32.4
COND:0-1-WT-S-X-I-NO-TD

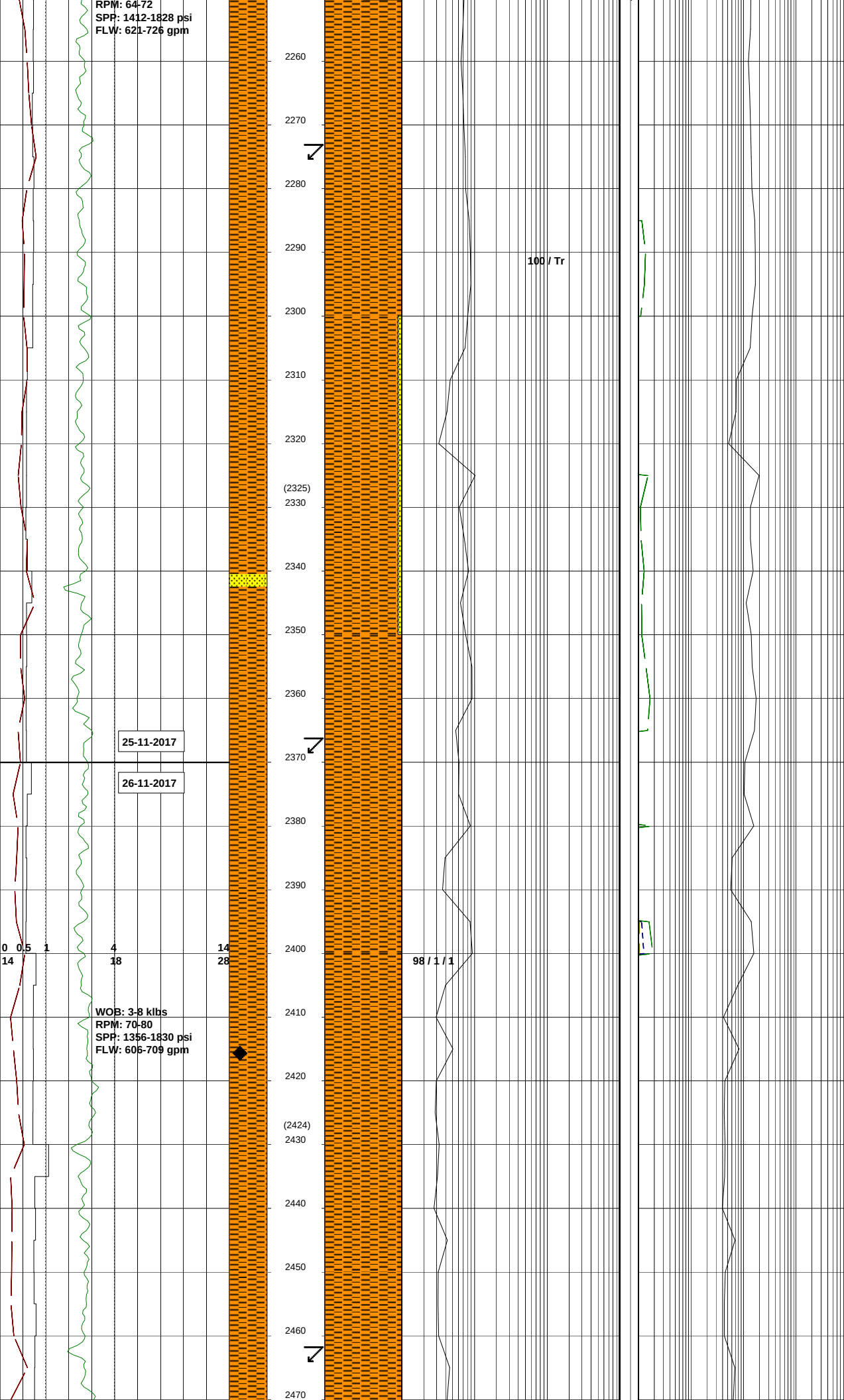
SILTSTONE:lt-med gry,off wh-v lt
gry,arg,com aren grd to vf SST,
calc,occ carb flks,frm-mod hd,
sbblky-sbflss.

SANDSTONE:off wh-lt gry,vf com
grd to aren SLTST,mod srt,sbrnd,
abdt mod calc cmt,com off wh arg
mtx,tr liths,mod hd,ti vis por,
no fluor.

OODNADATTA FORMATION:
2234'MDRT (-2084'TVDSS)

WOB: 1-13 kips

RPM: 64-72
SPP: 1412-1828 psi
FLW: 621-726 gpm



SURVEY @ 2273.04'MDRT
INC: 1.84° AZI: 280.01°G

SILTSTONE: med brn gry, olv gry,
lt olv gry-lt brnsh gry, arg, tr
aren, rr carb flks, frm-tr mod hd,
sbblky-sbfiss.

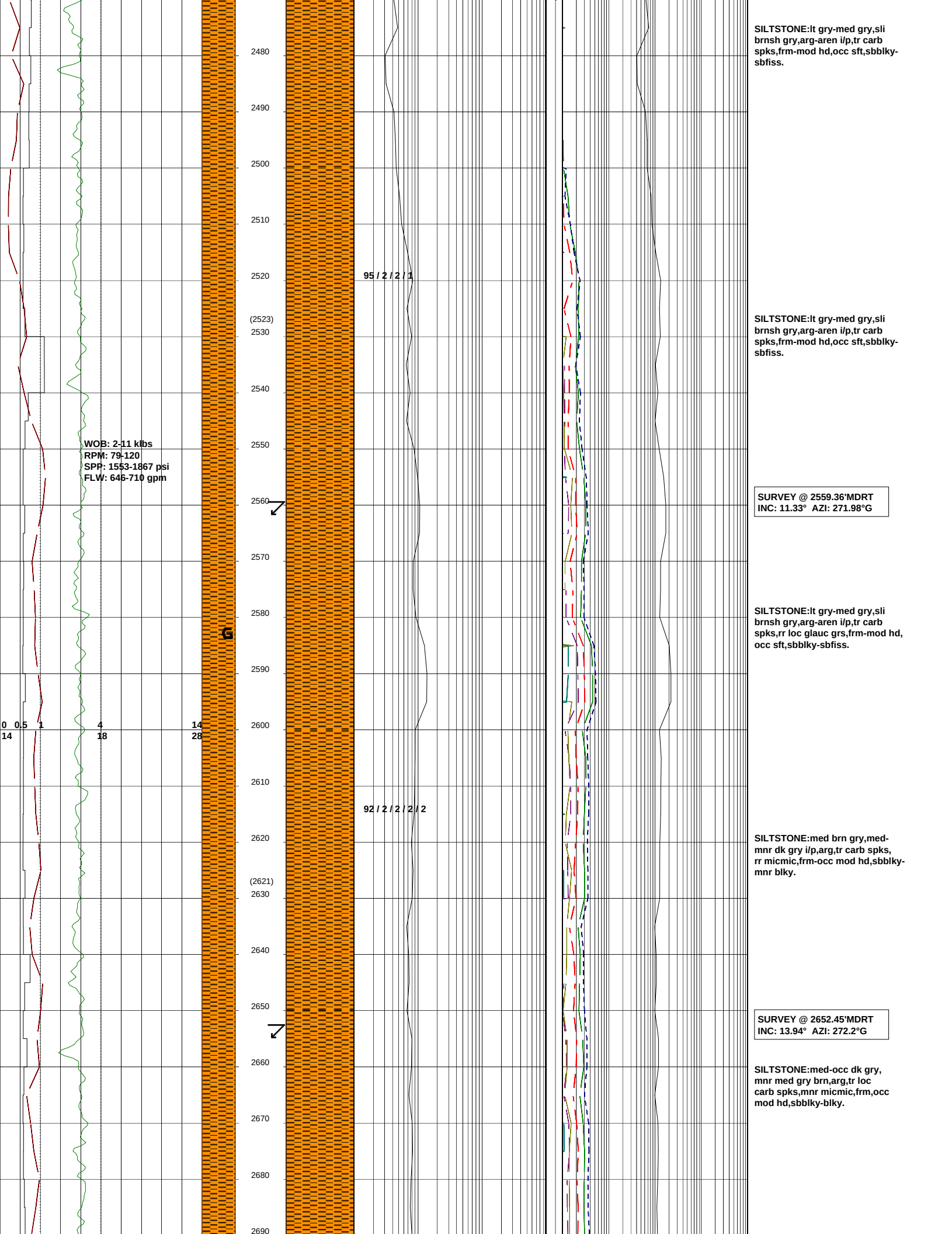
SANDSTONE: off wh-lt gry, vf com
grd to aren SLTST, mod srt, sbrnd,
tr mod calc cmt, com off wh arg
mtx, mnrr carb flks, tr liths, mod hd,
ti vis por, no fluor.

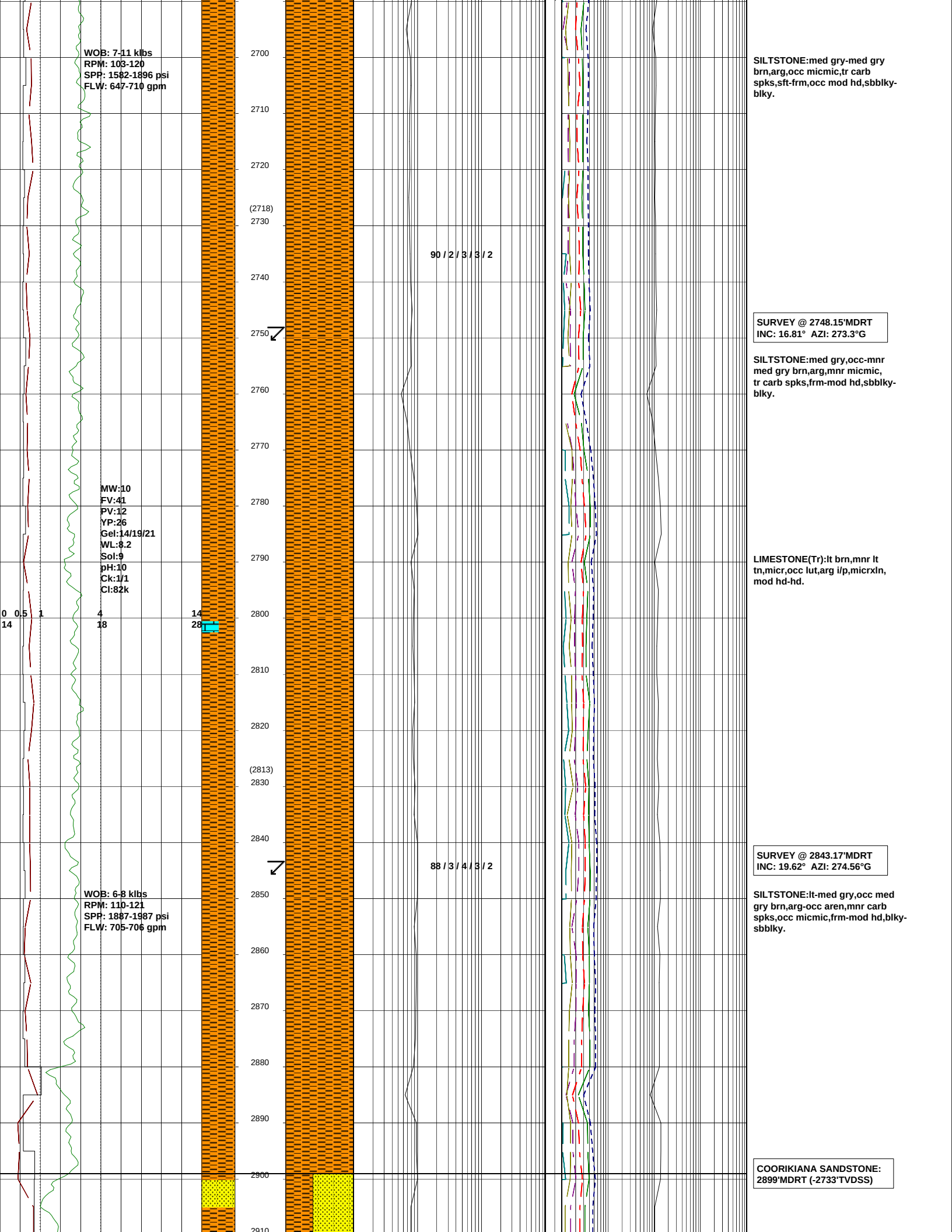
SURVEY @ 2366.16'MDRT
INC: 4.7° AZI: 267.73°G

SILTSTONE: lt gry-med gry, sli
brnsh gry, arg-aren i/p, tr carb
spks, frm-occ mod hd, sbblky-
sbfiss.

SILTSTONE: lt gry-med gry, sli
brnsh gry, arg-aren i/p, rr dissem
pyr, tr carb spks, frm-occ mod
hd, sbblky-sbfiss.

SURVEY @ 2461.7'MDRT
INC: 7.89° AZI: 269.46°G





WOB: 7-11 klbs
RPM: 103-120
SPP: 1582-1896 psi
FLW: 647-710 gpm

SILTSTONE: med gry-med gry
brn, arg, occ micmic, tr carb
spks, sft frm, occ mod hd, sbblky-
biky.

90 / 2 / 3 / 3 / 2

SURVEY @ 2748.15' MDRT
INC: 16.81° AZI: 273.3° G

SILTSTONE: med gry, occ-mnr
med gry brn, arg, mnr micmic,
tr carb spks, frm-mod hd, sbblky-
biky.

MW: 10
FV: 41
PV: 12
YP: 26
Gel: 14/19/21
WL: 8.2
Sol: 9
pH: 10
Ck: 1/1
Cl: 82k

LIMESTONE (Tr): lt brn, mnr lt
tn, micr, occ lut, arg i/p, micrxln,
mod hd-hd.

0 0.5 1 4 14
14 18 28

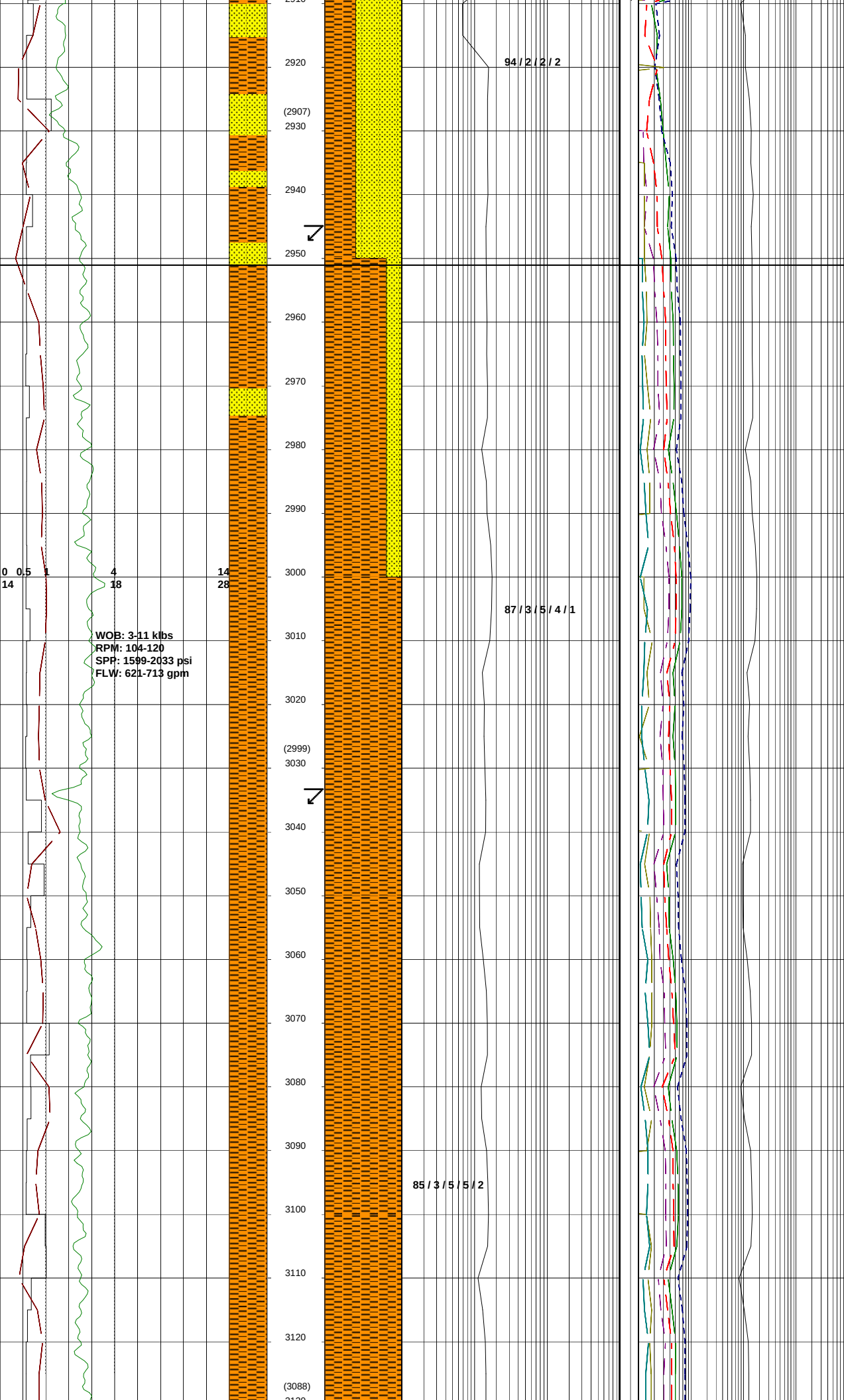
88 / 3 / 4 / 3 / 2

SURVEY @ 2843.17' MDRT
INC: 19.62° AZI: 274.56° G

SILTSTONE: lt-med gry, occ med
gry brn, arg-occ aren, mnr carb
spks, occ micmic, frm-mod hd, biky-
sbblky.

WOB: 6-8 klbs
RPM: 110-121
SPP: 1887-1987 psi
FLW: 705-706 gpm

COORIKIANA SANDSTONE:
2899' MDRT (-2733' TVDSS)



SANDSTONE:lt gry-lt grn,off wh i/p,vf-f,wi srt,sbang-sbrnd, mod calc cmt,com-abdt off wh-lt gry arg mtx,com glauc grs,fri-dom mod hd,v pr-ti vis por.

SURVEY @ 2938.99'MDRT
INC: 22.45° AZI: 274.78°G

BULLDOG SHALE:
2951'MDRT (-2781'TVDSS)

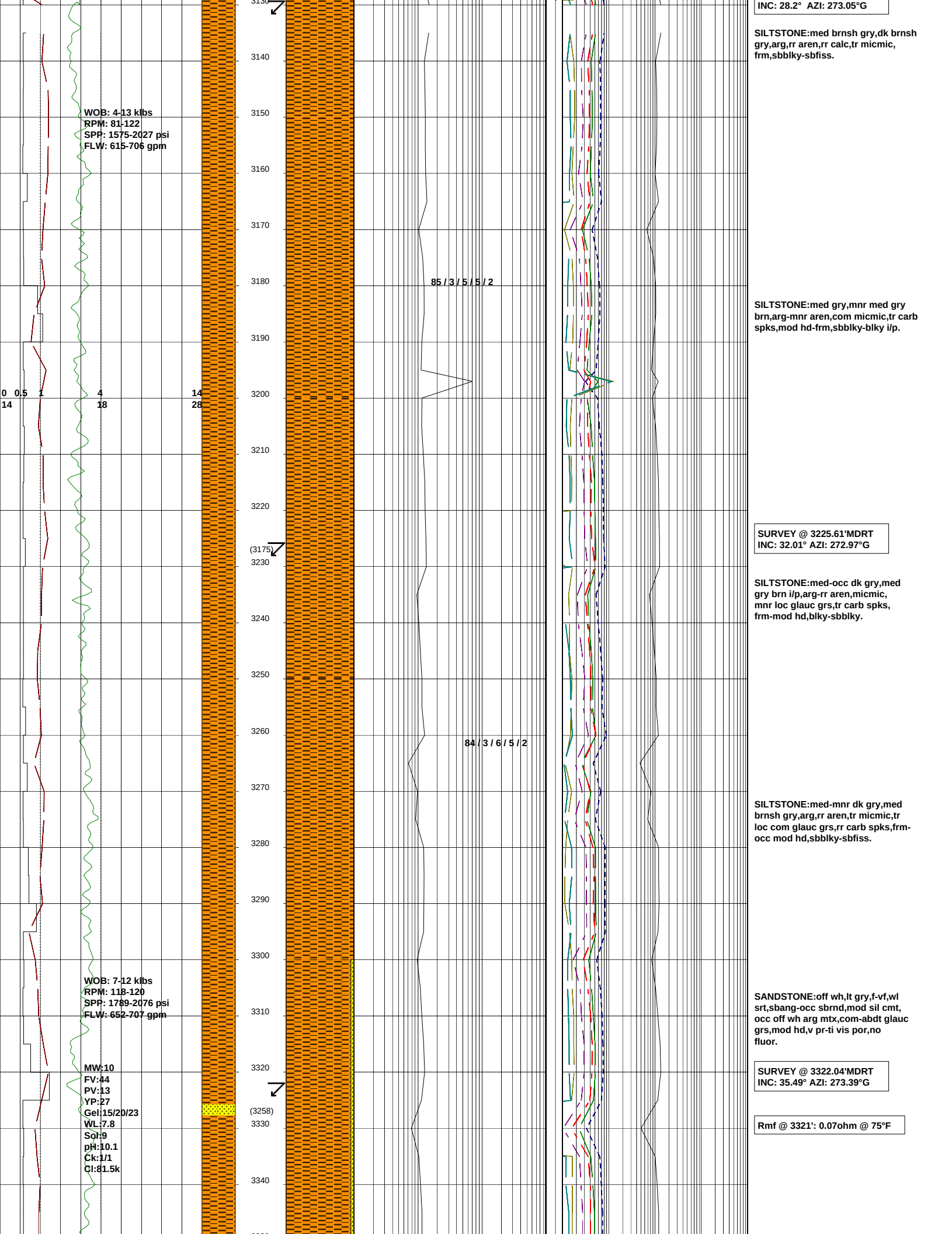
SILTSTONE:pl gry-med gry, pl brn gry,arg,aren i/p,tr carb spks,rr micmic,frm-mod hd, sbblky-blky.

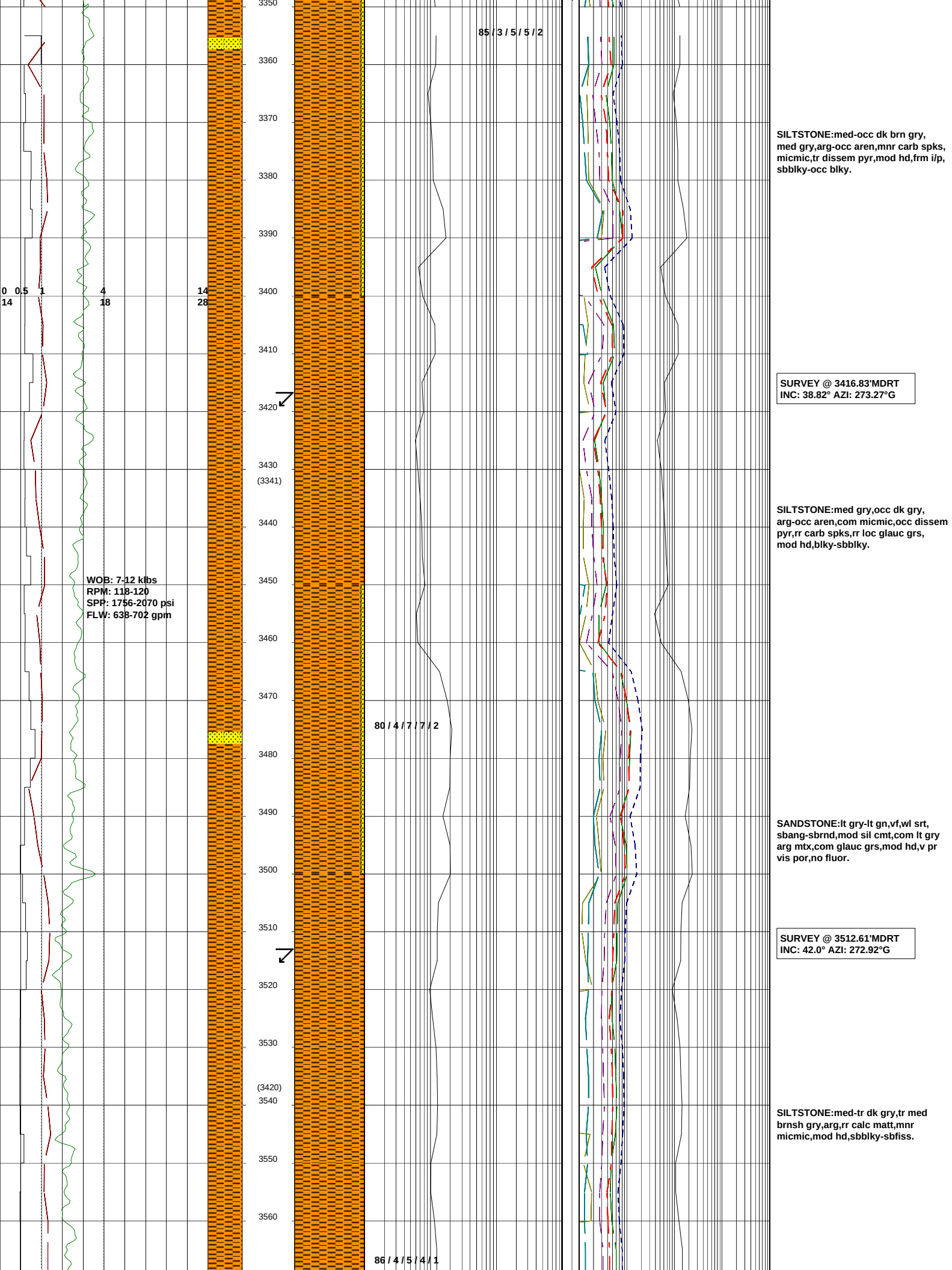
SILTSTONE:med gry-occ dk gry,pl brn gry,arg,aren i/p,tr carb spks,rr micmic,frm-mod hd,sbblky-blky.

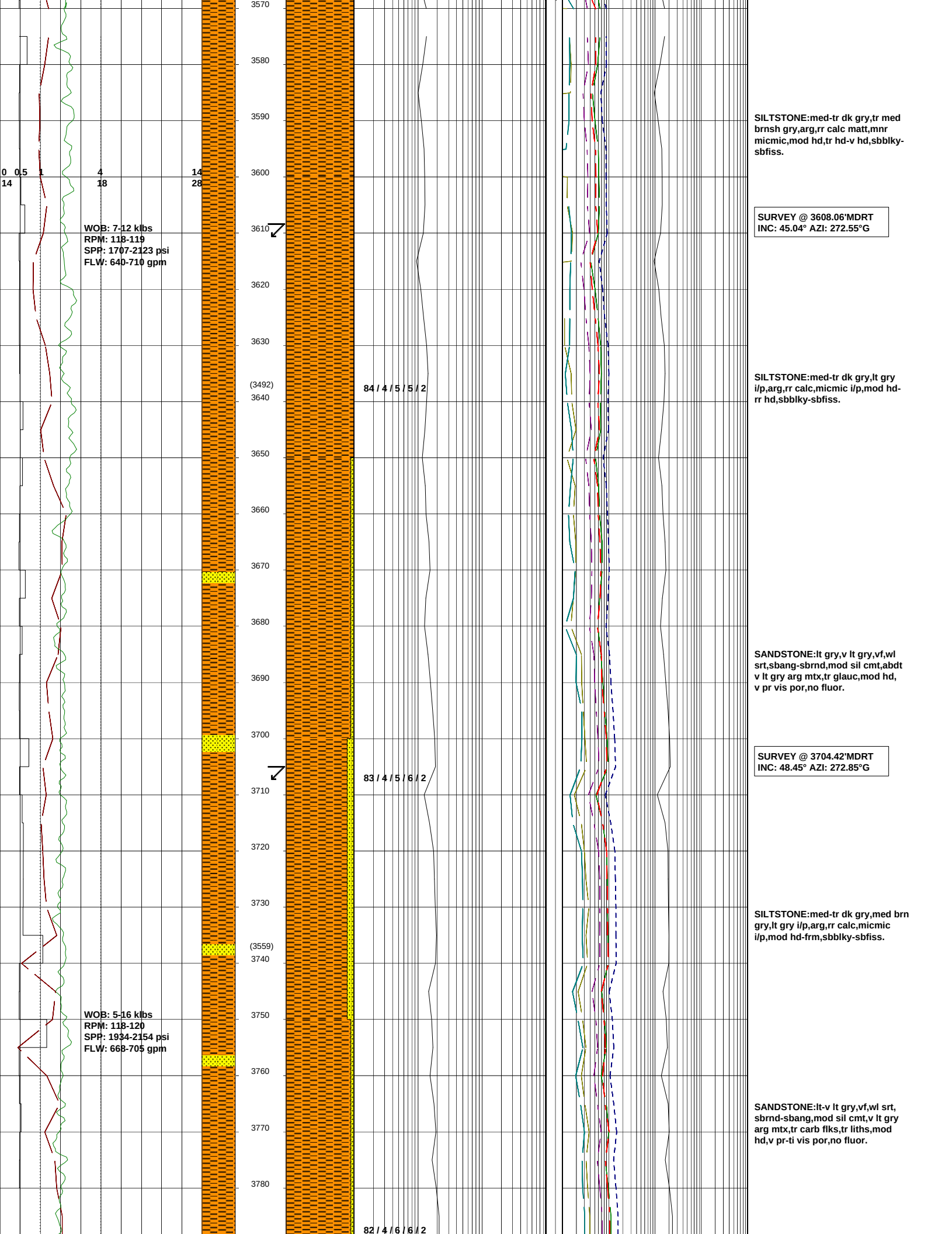
SURVEY @ 3033.74'MDRT
INC: 25.07° AZI: 274.61°G

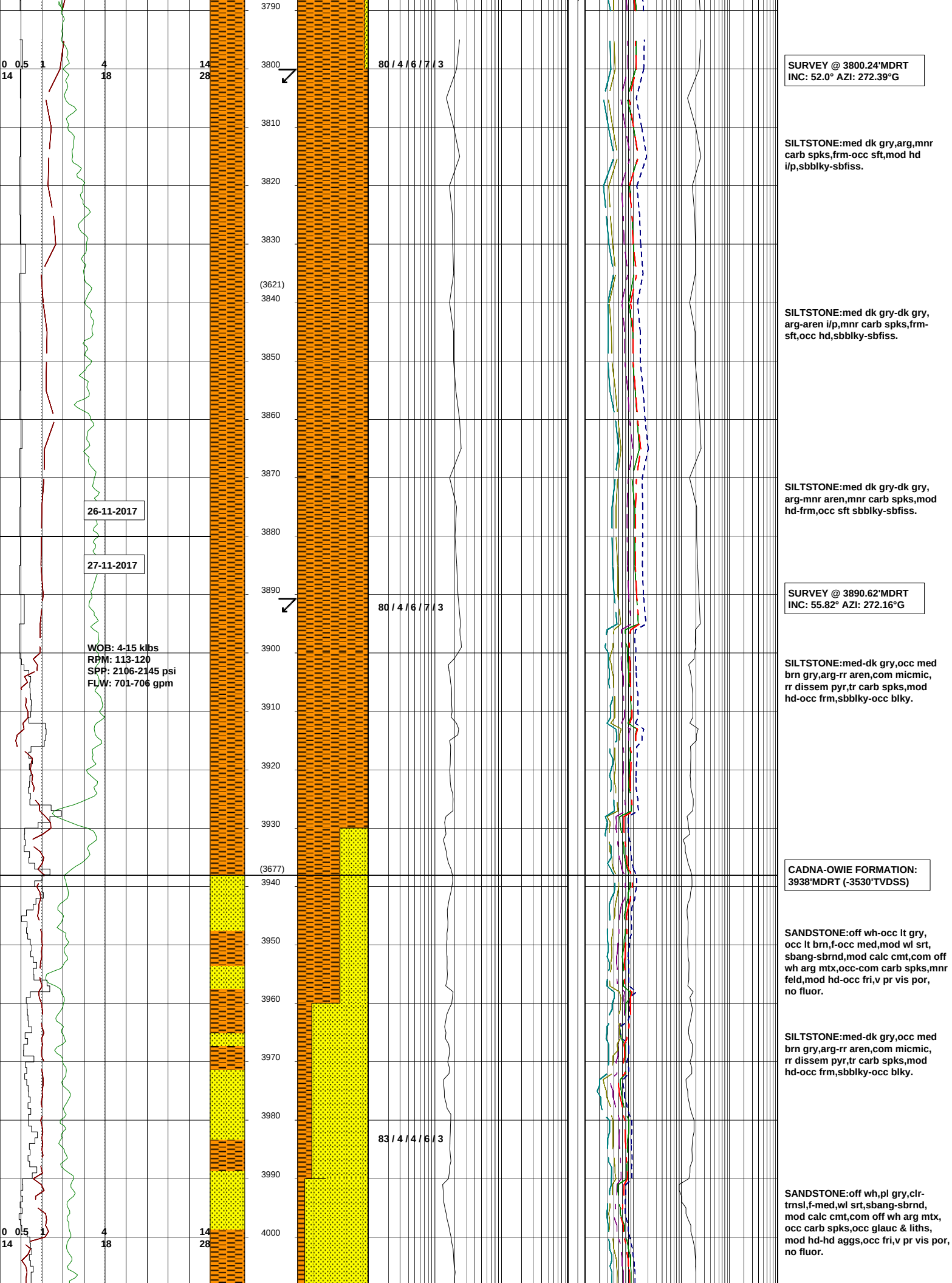
SILTSTONE:med gry-occ dk gry,pl brn gry,arg,aren i/p,tr carb spks,rr micmic,frm-mod hd,sbblky-blky.

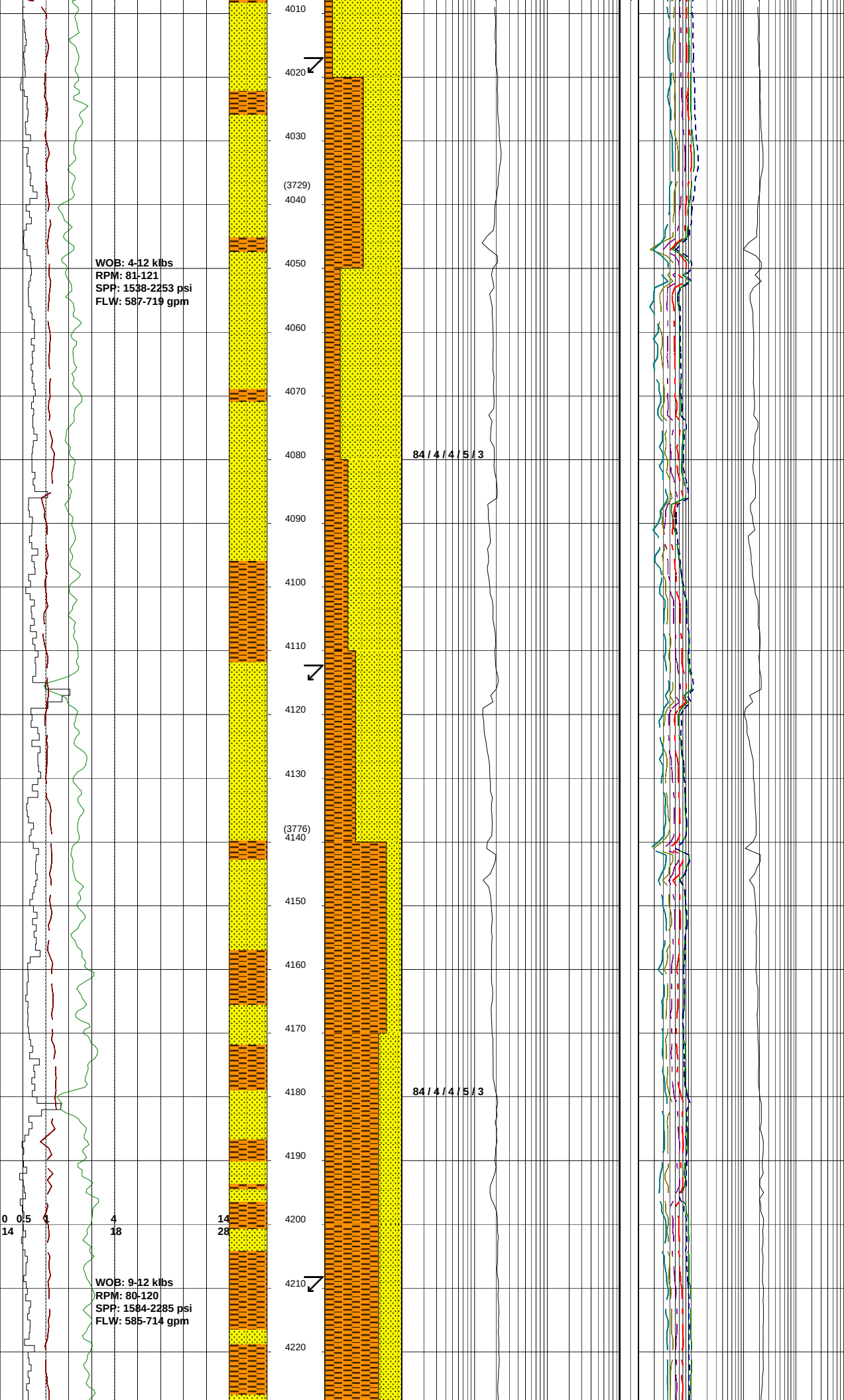
SURVEY @ 3129.89'MDRT











WOB: 4-12 klbs
RPM: 81-121
SPP: 1538-2253 psi
FLW: 587-719 gpm

WOB: 9-12 klbs
RPM: 80-120
SPP: 1584-2285 psi
FLW: 585-714 gpm

SURVEY @ 4016.95'MDRT
INC: 60.07° AZI: 271.96°G

SILTSTONE:med-dk gry,occ med
brn gry,arg-rr aren,com micmic,
rr dissem pyr,tr carb spks,mod
hd-occ frm,sbbiky-occ blkly.

SILTSTONE:med gry-dk gry,arg-
aren,tr carb spks,hd-occ mod
hd,sbbiky-blky.

SANDSTONE:off wh,pl gry,pl
brn,clr-trnsi,f-med,wl srt,sbang-
sbrnd,mod calc cmt,com off wh
arg mtx,occ carb spks,occ glauc &
liths,mod hd-hd aggs,occ fri,v
pr vis por,no fluor.

SURVEY @ 4112.18'MDRT
INC: 63.22° AZI: 271.44°G

SANDSTONE:off wh,pl gry,pl
brn gry,f-vf,occ med,wl srt,
rnd-sbrnd,strg calc cmt,com wh
arg mtx,occ carb spks & lams,com
liths,hd-mod hd,pr vis por,no
fluor.

SANDSTONE:off wh,pl gry,pl
brn gry,f-vf,occ med,wl srt,
rnd-sbrnd,strg calc cmt,com wh
arg mtx,occ carb spks & lams,
com liths,hd-mod hd,pr vis por,no
fluor.

SILTSTONE:lt-med brn,med-
occ dk brn gry,mnr dk gry,arg-
dom aren,occ carb spks,micmic,
mod hd-hd,sbbiky-occ blkly.

SURVEY @ 4208.17'MDRT
INC: 66.16° AZI: 271.76°G

SANDSTONE:lt brn-occ lt gry,
rr off wh,f,occ med-vf,mod srt,
sbang-occ sbrnd,mod calc cmt,
occ off wh-lt gry arg mtx,occ
feld & carb spks,mnr liths,hd,v
pr vis por,no fluor.

